

Use Case LM-1.2: Initialize Launch Platform

Scope: Launch Mission

Primary Actor: Flight Director

Secondary Actor: None

Related use Cases: *Use Case 1 Initialize Launch,*

Use Case 1.1 Initialize Rocket

Use Case 1.2.1 Initialize Launch Mechanism

Stake Holders & Interests:

- Flight Director: Wants to set up a Launch via initializing all the Systems and Sub-Systems including:
 - Launch Platform
- Mission Planner: Wants the Mission initialized with the appropriate Mission Parameters loaded

Pre-Conditions: The System is running and ready to initialize and monitor the appropriate Systems and Sub-Systems

Post-Conditions: The Launch Platform is initialized with the appropriate parameters. The Launch Platform is aware of the Rocket Initialization Status

Flight Director	System
1. Enters the Mission Parameters File (See <i>Use Case 1 Initialize Launch</i>)	
	2. Opens the Mission Parameters File
	3. Finds the Launch Platform Data
	4. Stores the Launch Platform Data
	5. Initialize each Launching Mechanism on the Launch Platform (See <i>Use Case 1.1.1 Initialize Launch Mechanism</i>)
	6. Query each Launching Mechanism for their Initialization Data
	7. Accesses the initialized Rocket Data (See <i>Use Case 1.1 Initialize Rocket</i>)
	8. Weighs the Rocket
	9. Returns the Initialized Launch Platform Data
10. Receives the Initialized Launch Platform Data	

Extensions (or Alternative Flows)

- 2a. If the System cannot open the Mission Parameters file, then the System alerts the Flight Director of the error. Flight Director determines next course of action.

- 3a. If the System cannot access the Launch Platform Data, then the System alerts the Flight Director of the error. Flight Director determines next course of action.
- 5a.,6a. If the System receives an error from any of the Launch Mechanisms (See *Use Case 1.2.1 Initialize Launch Mechanism*), then the System alerts the Flight Director of the error. Flight Director determines next course of action.
- 7a. If the System cannot access the Initialized Rocket Data (See *Use Case 1.1 Initialized Rocket*), then the System alerts the Flight of the issue. The Flight Director determines next course of action.
- 8a. If the System receives an error while weighing the Rocket, then the System alerts the Flight Director of the issue. The Flight Director determines the next course of action.

Notes/Constraints

- *a This Use Case documents the Launch Platform Initialization , not the Platform Monitoring (See *Use Case 4.1 Monitor Initialization*)
- *b Upon the receipt of an alert, the next course of action for the Flight Director is aborting the mission or chosing another Mission Parameter File.

Special Requirements:

- None

Technology & Variations List:

- There could be many types of Mission Parameter Files in different formats for the System Composition. The System needs to account for all the different types of files and read them accordingly.
- Regardless of the type of file, the System Data will follow a Standard

Frequency of Occurence

- Every time a Mission is planned or simulated

Open Issues

- 7b. There are different possiblities if there is no Initialized Rocket Data (See *Use Case 1.1 Initialized Rocket*). Possibly considering giving the Flight Director the option to Initialize. Possibly consider Initializing the System without a Rocket on the Platform.